MATH329 · OPERATIONS RESEARCH

Formulation Practice

Six formulation problems from the slides and the Examples file

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01 Questions

For each problem write the variables, then the objective function, then the constraints. Do not solve the problem. The full solution is in a separate file.

Q1 MAX · BADER

Bader manufactures men's shirts and women's blouses. It employs 25 workers in the cutting department, 35 in sewing and 5 in packaging. The factory works one 8-hour shift, 5 days a week. A shirt needs 20 minutes of cutting, 70 of sewing and 12 of packaging, with a profit of $8. A blouse needs 60, 60 and 4 minutes, with a profit of $12. Formulate the problem to find the optimal weekly production.

Q2 MAX · WILD WEST

Wild West produces two types of cowboy hats. Type I hat requires twice as much labor as Type II. If all the available labor time is dedicated to Type II alone, the company can produce a total of 400 Type II hats a day. The market limits are 150 and 200 hats per day for the two types. The profit is $8 per Type I hat and $5 per Type II hat. Formulate the problem as an LPP so as to maximize the profit.

Q3 MIN · FEED MIX

The manager of a milk dairy decides that each cow should get at least 15, 20 and 24 units of nutrients A, B and C respectively. Two varieties of feed are available. In feed of variety 1 (variety 2) the contents of the nutrients A, B and C are respectively 1 (3), 2 (2), 3 (2) units per kg. The costs of varieties 1 and 2 are respectively 2 Rs. and 3 Rs. per kg. How much of feed of each variety should be purchased to feed a cow daily so that the expenditure is least?

Q4 MIN · TONICS

A doctor advises a patient to consume at least 40 units of vitamin A and 50 units of vitamin D daily. Tonic X contains 2 units of vitamin A and 3 units of vitamin D per unit, and costs 5 Rs. per unit. Tonic Y contains 4 units of vitamin A and 2 units of vitamin D per unit, and costs 3 Rs. per unit. Formulate an LPP to minimize the cost of the tonics.

Q5 MAX · REDDY MIKKS

Reddy Mikks produces interior and exterior paints from two raw materials, M1 and M2. A ton of interior paint uses 6 tons of M1 and 1 ton of M2. A ton of exterior paint uses 4 tons of M1 and 2 tons of M2. The daily availability is 24 tons of M1 and 6 tons of M2. The profit per ton is $5000 for interior and $4000 for exterior paint. The daily demand for interior paint cannot exceed that for exterior paint by more than 1 ton, and the maximum daily demand for interior paint is 2 tons. Formulate the problem.

Q6

MIN · TRIM LOSS

A company has to manufacture the circular tops of cans. Two sizes, one of diameter 10 cm and the other of diameter 20 cm, are required. They are to be cut from metal sheets of dimensions 20 cm by 50 cm. The requirement of the smaller size is 20,000 and of the larger size is 15,000. Formulate the problem as an LPP so that the number of sheets used is a minimum.